



UNIVERSITY OF SCIENCE AND TECHNOLOGY (UST) · KIMS CAMPUS

Where **Artificial Intelligence** Meets **Materials & Manufacturing**

AIMAP Lab – AI for Materials & Processing · UST 소재·공정 인공지능 연구실

We build AI that designs metals, reads microstructures, predicts manufacturing processes, and runs autonomous experiments – bridging deep learning, physics-based simulation, and robotics to accelerate materials innovation.

RESEARCH AREAS



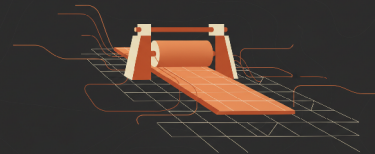
01 AI for Materials Design

Deep-learning property prediction and generative models that accelerate the design and discovery of new alloys.



02 AI-Based Materials Analysis

AI-driven microstructure characterization, automated data analysis, and ML-based performance evaluation.



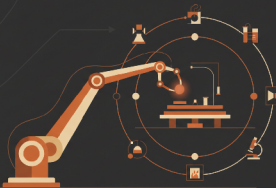
03 Physics-Informed AI for Processes

Hybrid AI-simulation frameworks that embed physical constraints for accurate, reliable process prediction.



04 Materials VLMs & AI Agents

Vision-language models and AI agents specialized for materials science that automate research workflows.



05 Autonomous Laboratory

Robotics + AI experimental platforms for closed-loop discovery and optimization of metallic materials.



06 Autonomous Manufacturing

Vision- and signal-based anomaly detection and physical AI for manufacturing systems that sense, decide, and act.

77

JOURNAL ARTICLES

61

CONFERENCE TALKS

38

FUNDED PROJECTS

3

OPEN-SOURCE TOOLS

PRINCIPAL INVESTIGATOR

**Ho Won Lee, Ph.D.** 이호원**Director**, Division of Materials AI Research, Korea Institute of Materials Science (KIMS)**Professor**, University of Science and Technology (UST)

Ph.D. Mechanical Engineering, KAIST · formerly Max-Planck-Institut für Eisenforschung (Germany)

Editorial Board, *Discover Metals* (Springer Nature) · Editorial Board, KSTP

Team: 1 Postdoc · 3 Ph.D. · 1 M.S. · 1 Intern

Crystal Plasticity + GNN

Diffusion Models

Self-Supervised Learning

VLM for Materials

SELECTED PUBLICATIONS – PI AS LEAD/CORRESPONDING AUTHOR

IF 11.4
top 1.9%**Quantifying microstructure- and deformation-mode-dependent heterogeneity using crystal plasticity simulations**
International Journal of Mechanical Sciences (2026) · doi.org/10.1016/j.ijmecsci.2026.111818IF 11.5
top 2.0%**Parameter-aware high-fidelity microstructure generation using stable diffusion**
Advanced Engineering Informatics (2026) · doi.org/10.1016/j.aei.2025.104080IF 9.0
top 3.1%**Synthetic microstructure generation using diffusion transformers with structure-preserving augmentation**
Engineering Applications of Artificial Intelligence (2026) · doi.org/10.1016/j.engappai.2026.114543IF 6.2
top 6.8%**Microstructure-informed forming limits in aluminum alloys: Experiments and crystal plasticity simulations**
Materials Characterization (2025) · doi.org/10.1016/j.matchar.2025.115831IF 8.2
Q1**Multi-fidelity physics-guided graph neural networks for structure-property relationships in polycrystalline metals**
Materials & Design (2026) · doi.org/10.1016/j.matdes.2026.116310IF 8.2
Q1**Multi-fidelity learning-based latent diffusion model for three-dimensional inverse microstructure design of dual phase steels**
Materials & Design (2025) · doi.org/10.1016/j.matdes.2025.114623

FLAGSHIP PROJECTS

PI **Foundation-model AI service for automated design of metal materials & processes**

NST (MSIT) · 2026–2030

PI **Digital innovation platform for metal materials manufacturing**

KIAT (MOTIE) · 2022–2026

Co-PI **AI-enabled virtual engineering platform for aerospace metal additive manufacturing**

KIAT (MOTIE) · 2025–2029

OPEN-SOURCE SOFTWARE

MatSSL – self-supervised representation learning for metallographic image segmentation
github.com/duchieuphan2k1/MatSSL**SD3.5-DreamBooth-LoRA** – parameter-aware fine-tuning for controllable microstructure generation**Microstructure-SegNet** – unified semantic segmentation pipeline for microstructure validation

RESEARCH INFRASTRUCTURE

**GPU Cluster**
(18× A100 · 24× A6000)**Automatic Rolling Mill****Auto Tensile Machine****X-Ray Diffractometer****Collaborative Robot****Large-Format SLA Printer**

We produce process and property data in-house on the rolling mill, tensile machine, and XRD, automate experiments with the robot arm and SLA printer, and train AI models on our GPU cluster – from data generation to validation in one lab.

Join Us – 함께할 연구자를 찾습니다

We welcome **M.S./Ph.D. students (UST)**, **postdoctoral researchers**, and **interns** passionate about AI for materials and manufacturing. International applicants welcome – our team works in English.

소재·제조 AI 연구에 함께할 대학원생(UST 석·박사), 박사후연구원, 인턴을 모집합니다.

Prof. Ho Won Lee · h.lee@kims.re.kr · +82-55-280-3843

Korea Institute of Materials Science, 797 Changwondae-ro, Seongsan-gu, Changwon, Korea



aimat.kr